

# Capstone Project: Modeling with Calculus — Schedule

Calculus II · STM 1002 · Dorothy Dickmann · University of Austin · Spring 2026

| Date   | Day | Session    | In class                         | Project                                                          |
|--------|-----|------------|----------------------------------|------------------------------------------------------------------|
| May 13 | Wed | L17        | Risk, hedging, and project intro | Part 1 distributed                                               |
| May 15 | Fri | L18        | Part 1 debrief; Part 2 setup     | Part 1 due (hard copy, start of class) · Part 2 distributed      |
| May 18 | Mon | L19        | Part 2 debrief                   | Part 2 due (hard copy, start of class) · Parts 3 & 4 distributed |
| May 20 | Wed | L20        | Project work day                 | In-class work time; questions on Parts 3 & 4                     |
| May 21 | Thu | —          | —                                | Final report due to Populi by midnight                           |
| May 22 | Fri | Discussion | Graded in-class discussion       | Bring a hard copy of your report for your reference              |

**Checkpoints** are completion-graded. The final report contains answers to all parts; debriefs are an opportunity to correct and improve checkpoint work. Final reports and the in-class discussion are graded for quality.

*This project uses real market data for educational purposes only and does not constitute investment advice.*

# Capstone Project: Modeling with Calculus

Calculus II · STM 1002

Dorothy Dickmann · University of Austin · Spring 2026

---

*This project uses real market data for educational purposes only and does not constitute investment advice.*

This project has four parts. Each has a specific purpose: price a real option using the calculus tools from this course (Part 1), apply the model to a hedging decision (Part 2), interrogate its assumptions (Part 3), and apply it to a problem of your choosing (Part 4). The sequence is deliberate: you should understand a model before you criticize it, and criticize it before you generalize it. The process you use here applies any time you're modeling a real-world phenomenon, not just in finance.

---

## The Black-Scholes formula for a European call option:

$$C = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2),$$

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

| Symbol        | Name              | In plain terms                                                                        |
|---------------|-------------------|---------------------------------------------------------------------------------------|
| $C$           | Call price        | price of the option                                                                   |
| $S_0$         | Underlying level  | current level of the underlying asset (stock, index, commodity)                       |
| $K$           | Strike            | asset level at which the option pays off                                              |
| $T$           | Time to expiry    | time to expiration, in years                                                          |
| $r$           | Risk-free rate    | interest rate on a riskless investment over the same horizon (our discount rate)      |
| $\sigma$      | Volatility        | how much the underlying asset price moves, annualized; the one parameter you estimate |
| $\Phi(\cdot)$ | Normal CDF        | $\Phi(z) = P(Z \leq z)$ for standard normal $Z$                                       |
| $d_1$         | Moneyness input   | determines $\Delta$ — how sensitive the option price is to a move in SPX              |
| $d_2$         | Probability input | $\Phi(d_2)$ is the probability the option pays off at expiration                      |

**What is a European call option?** A call option gives the buyer the right (but not the obligation) to purchase the underlying asset at strike  $K$  at expiration (*A put option gives the right to sell at  $K$  at expiration*). “European” means exercise is only possible at expiration, not before. The payoff is  $\max(S_T - K, 0)$ : positive if  $S_T$  closes above  $K$ , zero otherwise (*no obligation to exercise*). Calls are used to speculate on upside, hedge short positions, or cap future purchase costs (as when an airline hedges jet fuel) (*puts serve as portfolio insurance against a price decline*).

---

## Part 1: Option Pricing, FTC, and Taylor Approximation

In Part 1 you will price a European call option on the S&P 500 index (ticker: SPX) using the Black-Scholes model. SPX is a basket of 500 large-cap U.S. companies. The contract has strike  $K = 7900$  and expires November 20, 2026, paying off only if the index closes above 7900 on that date.

**Checkpoint — Friday, May 15, hard copy:** (a) hand computations and answers to all parts below, organized and legible, and (b) a PDF export of your Python notebook.

**Project parameters.** Ticker: SPX261120C07900000 (SPX European call, strike 7900, November 20, 2026 expiry). Use these values throughout the project.

|                                |                                                                        |
|--------------------------------|------------------------------------------------------------------------|
| $S_0 = 7414$                   | SPX index level, in index points (closing price on data pull date)     |
| $K = 7900$                     | strike, in index points                                                |
| $T = 0.523$ yr                 | time to expiration (November 20, 2026 expiry)                          |
| $r = 0.036$                    | T-bill rate (May 13, 2026)                                             |
| $C_{\text{market}} = \$180.30$ | observed option price, in dollars (bid-ask midpoint on data pull date) |

*A note on units.*  $S_0$  and  $K$  are in index points;  $C$  is in dollars. This is consistent: by convention, one index point equals \$1, so the formula gives a dollar price for the option.

(a) **Estimate  $\sigma$  from historical data.**

Open the starter notebook at <https://tinyurl.com/uatxoptionspart1>. The notebook downloads historical SPX closing prices, computes daily log-returns  $r_t = \ln(S_t/S_{t-1})$ , and annualizes the sample standard deviation by  $\sqrt{252}$  to produce  $\sigma$ . Your one decision is the length of the historical window (`lookback_days`). Try a few values, think about the trade-off between stability and recency, and settle on one. Report your estimated  $\sigma$ , the number of returns used, and a brief rationale for your choice.

(b) **Compute  $\Phi(d_1)$  and  $\Phi(d_2)$  using the Taylor series.**

The standard normal CDF is defined as

$$\Phi(z) = \int_{-\infty}^z \varphi(t) dt, \quad \text{where } \varphi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}.$$

Your goal is to evaluate  $\Phi(d_1)$  and  $\Phi(d_2)$  numerically by integrating a Taylor series term by term. Follow the steps below.

- (i) Write the Taylor series for  $e^x$  centered at  $x = 0$ . Give the first five terms and the general term  $\sum_{n=0}^{\infty}(\dots)$ .
- (ii) Substitute  $x = -t^2/2$  to obtain a power series for  $e^{-t^2/2}$ . Write out the first five terms explicitly, simplifying each coefficient.

(iii) Use the series from step (ii) to integrate  $\varphi(t)$  and compute  $\Phi(z)$ . Express your answer in sigma notation.

(iv) First compute  $d_1$  and  $d_2$  using your  $\sigma$  from Part 1(a):

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

Then use your series from step (iii) to evaluate  $\Phi(d_1)$  and  $\Phi(d_2)$ . Use the next-term error bound to determine when to stop: add terms until the next term is smaller than 0.0001 in absolute value. State how many terms you needed for each.

(c) **Compute  $C$  by hand.**

Using your  $\Phi(d_1)$  and  $\Phi(d_2)$  from Part 1(b), compute the Black-Scholes call price:

$$C = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2).$$

Show every arithmetic step. Write the formula before substituting numbers.

(d) **Build a Python Black-Scholes function and verify.**

In the notebook, complete the `black_scholes_call(S0, K, r, sigma, T)` function using `scipy.stats.norm.cdf` for  $\Phi$ . A sanity check is already set up: your function should return  $C \approx 10.45$  for  $S_0 = K = 100$ ,  $r = 0.05$ ,  $\sigma = 0.20$ ,  $T = 1$ . Show the sanity check output.

Then run the function on the project parameters. Report two comparisons: (1) your Taylor  $\Phi(d_1)$  and  $\Phi(d_2)$  from Part 1(b) vs. `norm.cdf(d1)` and `norm.cdf(d2)` — they should agree to at least three decimal places; (2) your hand-computed  $C$  from Part 1(c) vs. the Python output — they should agree to within rounding.

(e) **Compare to the market price.**

The observed market price for this contract on the data pull date is  $C_{\text{market}} = \$180.30$ . Compare this to your model price  $C$  from Part 1(c). Is the market price higher or lower than your model predicts, and by how much? In 3–4 sentences, suggest what might account for the gap and explain your reasoning, drawing on class discussion, the course readings, or any additional research you find helpful.